

ERA5-Augmented Ensemble Learning for Multi-Horizon Tropical Cyclone Track Prediction over the Bay of Bengal: A Leakage-Aware Benchmark

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Abstract—Track guidance for Bay of Bengal tropical cyclones underpins evacuation timing along one of the world’s most exposed coastlines, yet full numerical weather prediction is costly for national agencies in the region. We benchmark eight tree-based track predictors at 3, 12, 24, and 48 h lead times on North Indian Ocean IBTrACS storms with genesis in 1940–2024, augmenting 49 kinematic features with 14 ERA5 reanalysis fields collocated at the storm centre. The proposed Subset-Expert Context-Aware Ensemble (SECE v2 Phase 3) routes horizon-specific subset experts, including a dedicated atmospheric “envi-ron” expert, through validation-only non-negative least squares fusion. We also report and correct a test-set leakage mode of our own making: architectural phases had been advanced while inspecting test-set rank tables, so we froze the design on nine development seeds and evaluated nine disjoint held-out seeds exactly once. On the held-out confirm (442 distinct test storms pooled), SECE attains median great-circle errors of 4.895, 35.538, 101.186, and 246.331 km. Paired Wilcoxon signed-rank tests with Bonferroni correction show SECE is significantly better than all seven baselines at 3 and 12 h, statistically indistinguishable from the leading stacks at 24 and 48 h, and never significantly worse than any baseline at any horizon. Ablations attribute long-lead gains to ERA5, while $5^\circ/3^\circ$ spatial patch statistics were tested and rejected. CNN-GRU and BLSTM are omitted from the locked confirm table after trailing tree-based systems in exploratory development under the same uniform-hyperparameter protocol [11].

Index Terms—Tropical cyclone track prediction, ERA5 reanalysis, ensemble learning, Bay of Bengal, IBTrACS, evaluation protocol, data leakage.

I. INTRODUCTION

The Bay of Bengal concentrates a disproportionate share of global tropical cyclone fatalities into a shallow, funnel-shaped basin bordered by densely populated deltaic coastline. Cyclone Sidr in 2007 and the storm-surge events that followed it drove sustained investment in early warning, for which track uncertainty is the quantity that sets evacuation lead time [3], [4], [13]. Operational guidance rests on dynamical models and consensus aids [9], [16], but the computational and assimilation infrastructure they require is not uniformly available to national meteorological agencies in the region.

Data-driven track models offer a complementary path. Best-track archives supply decades of labelled trajectories [1], [2], and global reanalysis supplies a physically consistent estimate of the atmospheric state at each storm position [24], so a

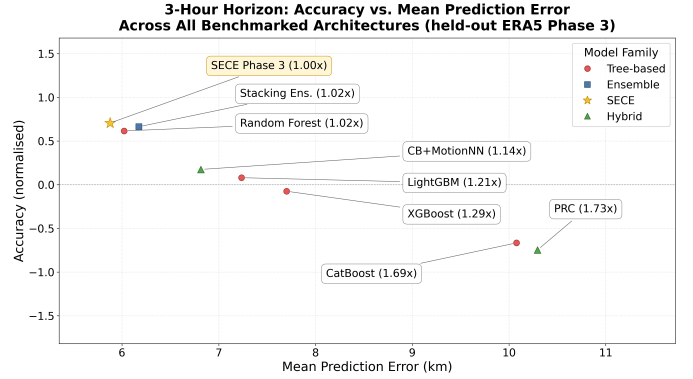


Fig. 1. Held-out 3 h benchmark (pooled nine seeds). Horizontal axis: mean great-circle error (km); vertical axis: normalised accuracy from median error vs. the tree family. Table II reports pooled medians with 95% bootstrap CIs.

learned model can consume environmental context without running a forecast cycle [7], [19]. Reported approaches span climatological-persistence analogues [12], [15], recurrent and convolutional sequence models [5], [6], [18], and transformer variants [20]. On engineered tabular data at the sample sizes available for a single basin, however, tree ensembles remain hard to displace [10], [11].

Three gaps motivate this work. First, basin-specific benchmarks for the North Indian Ocean typically compare few architectures, so relative claims are fragile [8], [17]. Second, evaluation is often confined to a single lead time, leaving horizon-dependent behaviour undocumented [19]. Third, the incremental value of reanalysis over pure kinematics is rarely isolated by controlled ablation. A fourth gap is methodological rather than architectural: when ensemble design is refined while the designer inspects test-set rankings, the reported held-out estimate is optimistic even though no model ever saw a test label [29].

This paper contributes an eight-system benchmark across four horizons under storm-wise splits, bootstrap confidence intervals, and paired significance testing; the SECE Phase 3 architecture with a dedicated ERA5 environ expert and validation-only routing; a documented correction of the leakage mode above, including the fact that our pre-correction results did not replicate; and a controlled ablation reporting

the rejection of spatial patch features as a negative result.

Deep sequence baselines (CNN-GRU, BLSTM) were scoped out after exploratory development under the same fixed budget as the trees (300 estimators/epochs, lr 0.01): both trailed tree and hybrid leaders at 3–24 h [11], so the locked confirm lists eight tree systems only (Table II).

II. DATA AND FEATURES

A. Archive, quality control, and splits

We use IBTrACS v4 North Indian Ocean best tracks [1], [2]. After quality control on 3-hourly rows, which enforces consistent timesteps and valid displacement targets, the modelling table contains 852 storms and 27,728 forecast origins. Multi-horizon supervision at 12, 24, and 48 h requires a sufficient forward track within each storm; 583 storms satisfy this and yield 15,605 multi-horizon origins. Because storm assignment is fixed once at the multi-horizon level, all four horizons draw from the same storm pool.

Partitioning is by storm identifier rather than by observation, so autocorrelation within a trajectory cannot cross partitions. Each seed draws a 70:15:15 storm-wise split, which yields 408 training, 87 validation, and 88 test storms per seed; sample counts vary with the length distribution of the storms drawn, for example 11,066 training and 2,370 test multi-horizon samples on seed 0.

B. ERA5 collocation

ERA5 fields [24] are extracted over 30°N–5°N, 70°E–102°E at 3-hourly resolution for every day on which a tracked storm was active, and joined to each track row. The 14 point features are zonal and meridional wind at 850, 500, and 200 hPa, mean sea level pressure, sea surface temperature, a derived deep-layer steering wind, and vertical shear expressed as 200 hPa minus 850 hPa components together with its magnitude. Because ERA5 sea surface temperature is undefined over land, near-coast rows are filled from the nearest valid ocean cell and flagged; the flag is retained as a feature so the model can condition on the substitution.

The 49 kinematic and derived predictors cover lagged positions, translation speed and bearing with sinusoidal encoding, acceleration, bearing curvature, seasonal encoding, a latitude-and-bearing recurvature proxy, distance to land, and landfall state, following established practice for this basin [5], [25].

The subset-expert design assumes these features carve the data into physically meaningful regimes, so we check that before modelling. Fig. 2 projects the 63-dimensional space with t-SNE [27], coloured independently by storm nature, translation speed, latitude band, and displacement magnitude. Coherent structure under all four colourings, not just one, indicates the feature space separates distinct kinematic and environmental regimes.

C. Genesis window

We compared genesis windows of 1940–2024, 1979–2024, and 1990–2024. Because observational quality improves sharply once satellite coverage begins, a naive comparison

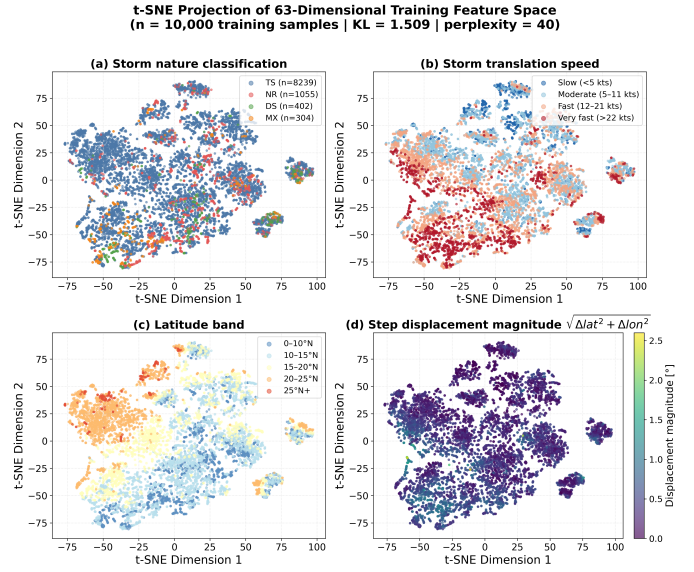


Fig. 2. t-SNE projection [27] of the 63-dimensional feature space ($n = 10,000$), coloured by independent meteorological attributes. Coherent separation indicates the engineered features encode physically distinct regimes, which is the premise underlying the subset-expert design.

confounds training-set size with test-set composition, so we repeated the comparison with all three models evaluated on one common modern holdout. The widest window won at every horizon on that fixed test, by 0.18 km at 3 h but by 7.9 km at 24 h and 17.7 km at 48 h relative to the modern-only model. We read this as position records remaining usable in earlier decades even where intensity observations are sparse, which suffices because the target here is track geometry rather than intensity, and we lock 1940–2024 accordingly.

III. VALIDATION PROTOCOL

Early development advanced SECE through successive architectural phases on a single fixed split, and the decision to advance from one phase to the next was taken by inspecting per-horizon test-set rank tables and adding fusion paths that closed the visible gaps. No test observation entered any fitting step, but the architecture was selected using test feedback, which is sufficient to invalidate the resulting held-out estimate [29]. The concrete symptom was that a design which appeared to win at every horizon on the original split did not reproduce that sweep on fresh splits.

We therefore separate two disjoint seed tiers. Nine development seeds {3, 4, 5, 6, 8, 9, 10, 11, 14} carry every architectural and fusion-path decision. Nine held-out seeds {0, 1, 2, 7, 13, 99, 123, 2024, 2026} were generated and evaluated exactly once, after the architecture was frozen. Convergence is verified by agreement between the tiers rather than by further tuning: at 48 h the development and held-out means differ by 0.05 km (246.94 against 246.99 km).

The primary metric is median great-circle error in kilometres, computed per test storm and aggregated [33]. The held-out confirm adds 95% confidence intervals from 10,000 storm-

level bootstrap resamples [30] and paired Wilcoxon signed-rank tests on matched per-storm medians [31], with Bonferroni correction across the seven simultaneous comparisons [32] giving a threshold of $\alpha = 0.05/7 \approx 0.0071$.

IV. METHOD

A. Baselines

Seven comparators span the tree and hybrid families: a Ridge-penalised stacking ensemble over Random Forest, XGBoost, and LightGBM base learners; a Persistence Residual Cascade (PRC) that corrects a persistence extrapolation with a LightGBM residual fitted in kilometre space; Random Forest [26]; LightGBM [22]; XGBoost [21]; CatBoost [23]; and CB+MotionNN, a CatBoost predictor corrected by a 64-32-16 multilayer perceptron acting on kinematic residuals. All tree learners share 300 estimators, maximum depth 6, and learning rate 0.01, so that differences reflect architecture rather than tuning budget [11].

B. SECE Phase 3

For each horizon, SECE builds four physics-informed feature subsets, shown in Fig. 3. At 3 h these are position, motion, environ, and the full set; at 12, 24, and 48 h they are motion, physics, environ, and full. Each subset is modelled by all four tree algorithms, giving 16 subset-level models per horizon, which are fused by non-negative least squares into one subset-consensus signal. That signal joins the seven baseline predictions in a second-stage fusion pool, from which horizon-specific candidates are formed: degree-space fusion over the full pool, a persistence-anchored kilometre-space variant, and at 3 h an additional stacking-plus-residual path. A hard router then selects, per seed, the candidate with the lowest *validation* median error, and that frozen choice produces the reported test predictions.

The environ expert is the locked architectural addition. Rather than dispersing the 14 ERA5 features into existing subsets, we allocate them, together with the native environmental fields, to a dedicated subset with its own four-algorithm pool. This gives atmospheric information explicit representation in the ensemble and was selected over the alternatives in Section V-A.

V. RESULTS

A. Development ablation

Table I reports, per configuration, the number of development seeds on which SECE ranked first among all eight systems, and its mean median error. Adding point ERA5 features reduces the 48 h mean from 250.26 to 246.80 km, with 8 of 9 seeds improving, which establishes that atmospheric information carries exploitable signal at long lead. Routing those features through a dedicated environ expert then improves 24 h specifically, from 99.86 to 99.43 km, which is the horizon at which the plain point-ERA5 configuration was weakest. The same table records an honest cost: at 3 h the track-only configuration retains the lowest mean, 4.84 against 4.93 km, so ERA5 does not help uniformly across horizons.

TABLE I
DEVELOPMENT ABLATION OVER NINE DEVELOPMENT SEEDS. EACH CELL GIVES SECE RANK-1 COUNT OUT OF NINE, THEN MEAN MEDIAN ERROR IN KM. THE LOCKED ROW IS IN BOLD; THE FINAL ROW IS THE INDEPENDENT HELD-OUT CONFIRM FOR REFERENCE.

Configuration	3 h	12 h	24 h	48 h
Track-only (no ERA5)	5/9, 4.84	3/9, 35.48	3/9, 100.48	0/9, 250.26
+ point ERA5	5/9, 4.93	7/9, 35.59	2/9, 99.86	1/9, 246.80
+ environ expert	5/9, 4.93	6/9, 35.56	3/9, 99.43	2/9, 246.94
+ spatial 5°	7/9, 4.93	7/9, 35.56	1/9, 99.74	1/9, 246.55
+ spatial 3°	7/9, 4.93	7/9, 35.66	2/9, 99.42	3/9, 247.00
Held-out confirm (locked)	6/9, 4.90	6/9, 35.63	2/9, 101.21	1/9, 246.99

Spatial patch statistics at 5° and 3° raise short-range rank-1 frequency but leave 24 and 48 h means within roughly 0.4 km of the environ configuration, inside the seed-to-seed noise. We therefore did not adopt them, and report the rejection rather than omitting it.

B. Held-out confirm

Table II summarises pooled held-out medians and bootstrap intervals for all eight tree systems. SECE leads at 3 and 12 h (4.895 and 35.538 km), while stacking/XGBoost and XGBoost/PRC take the top two ranks at 24 and 48 h; confidence intervals overlap heavily at those longer leads. Fig. 1 shows the same 3 h ordering on a mean-error axis (caption: Table II uses medians), where the three leading systems separate clearly from the remainder.

Paired Wilcoxon signed-rank tests on 442 matched per-storm medians, with Bonferroni correction across seven simultaneous comparisons ($\alpha \approx 0.0071$) [31], [32], translate those ranks into significance. At 3 and 12 h, SECE is significantly better than every baseline after correction. At 24 h, only CatBoost and CB+MotionNN remain significant; Random Forest is significant before correction ($p = 0.0105$) but not after. At 48 h, only CatBoost remains significant after correction, while stacking and CB+MotionNN are significant before correction ($p = 0.037$) but not after. Every rank-biserial correlation is negative, so SECE is never significantly worse than any baseline at any horizon even where it does not top the median table. Counting rank-1 seeds reinforces the same pattern: SECE leads on 6 of 9 held-out seeds at both 3 and 12 h, but only 2 of 9 at 24 h and 1 of 9 at 48 h, where PRC and XGBoost win more often even though pooled medians remain close.

C. Where ERA5 actually helps

To locate ERA5's contribution independently of ensemble machinery, we compare a single LightGBM model trained with and without ERA5 features, storm by storm, on one development seed with 88 test storms. At 24 h the two variants are at rough parity, with 43 of 88 storms improved and a near-zero mean effect. At 48 h 54 of 88 storms improve, with a median gain of 10.9 km. The benefit therefore strengthens with lead time, consistent with a storm's own recent kinematics becoming a progressively weaker predictor of its future position while large-scale steering becomes more informative [7], [9].

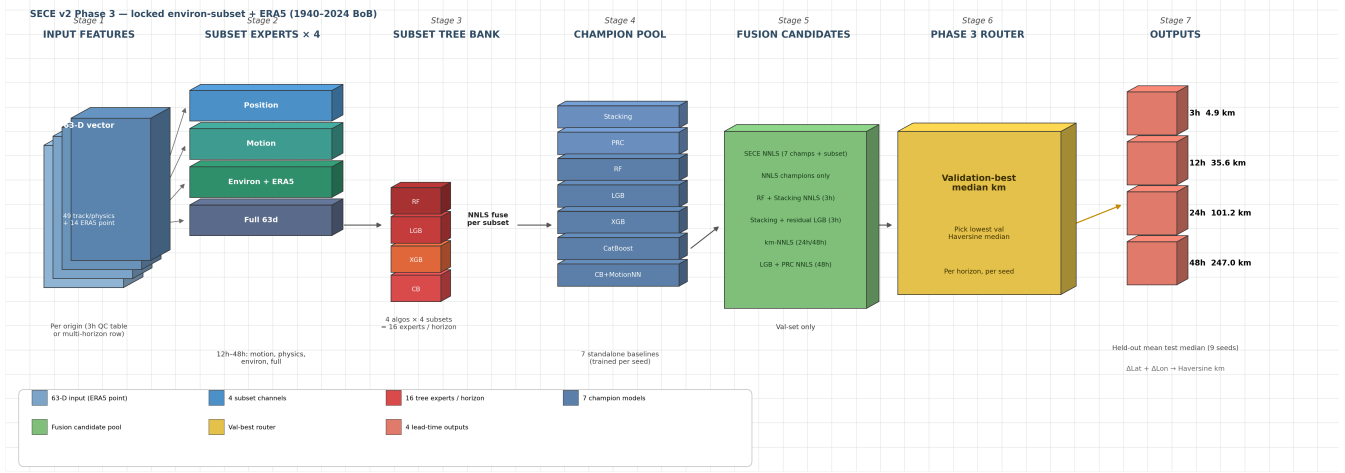


Fig. 3. Locked SECE Phase 3 pipeline. Sixty-three inputs, 49 kinematic and 14 ERA5 point features, feed four horizon-specific subset experts, each modelled by four tree algorithms and fused by non-negative least squares; a validation-only router selects the final fusion candidate per seed.

TABLE II

HELD-OUT TEST PERFORMANCE RANKED AT EACH HORIZON: MODEL AND MEDIAN GREAT-CIRCLE ERROR (KM) WITH 95% BOOTSTRAP CONFIDENCE INTERVAL. NINE HELD-OUT SEEDS POOLED (442 DISTINCT STORM IDENTIFIERS; 34,114 TEST SAMPLES AT 3 h AND 21,442 AT 12–48 h). EIGHT TREE/HYBRID SYSTEMS ONLY; CNN-GRU AND BLSTM OMITTED (SEE SECTION I).

Rank	3 h Horizon			12 h Horizon			24 h Horizon			48 h Horizon		
	Model	Med.	95% CI	Model	Med.	95% CI	Model	Med.	95% CI	Model	Med.	95% CI
1	SECE	4.895	4.71–5.08	SECE	35.538	34.2–36.9	Stacking	100.771	96.9–104.7	XGBoost	242.702	232.6–255.4
2	RF	4.927	4.73–5.13	Stacking	36.146	34.7–37.7	XGBoost	101.029	96.9–105.3	PRC	243.229	231.7–255.2
3	Stacking	4.945	4.76–5.12	RF	36.901	35.5–38.3	SECE	101.186	97.3–104.8	LGB	245.975	234.9–257.2
4	CB+MoNN	5.813	5.70–5.93	LGB	36.928	35.5–38.3	LGB	101.632	97.8–105.6	Stacking	245.977	235.9–257.2
5	LGB	6.026	5.89–6.17	XGB	36.947	35.5–38.4	PRC	101.941	98.1–105.8	SECE	246.331	235.5–257.6
6	XGB	6.411	6.24–6.58	PRC	37.177	35.7–38.5	RF	103.511	99.7–107.4	CB+MoNN	249.678	239.0–261.2
7	CatBoost	8.479	8.28–8.69	CB+MoNN	37.413	36.0–38.9	CB+MoNN	104.034	100.2–108.1	RF	251.305	240.8–262.3
8	PRC	8.872	8.71–9.06	CatBoost	42.028	40.4–43.7	CatBoost	107.191	102.6–111.8	CatBoost	254.862	243.0–267.8

Gain-based importances within the environ expert (held-out seed 0 only; see Limitations) reinforce this reading: kinematics still dominate, while SST, 200 hPa wind, and MSLP rank highest among ERA5 fields, with shares rising from 24 to 48 h. Shear and steering are collinear by construction, so gain may split arbitrarily among them; we report only the lead-time growth of ERA5 share, not within-group rank order.

D. Case study

Fig. 4 shows Cyclone Laila (2010), the storm with the largest single-storm ERA5 benefit in the diagnostic above: 25.0 km at 24 h and 107.0 km at 48 h. At the ensemble level on this storm SECE reaches 111.3 km at 24 h against 128.9 km for the pooled-best competitor, while at 48 h the ordering reverses and XGBoost leads locally, 204.9 against 236.1 km, mirroring the aggregate leaderboard. A representative storm chosen because its SECE error sits closest to the seed median, an unnamed 1996 system with central-basin genesis, shows a tighter contest at 24 h, 107.2 against 110.7 km. Per-storm behaviour therefore illustrates the mechanism but does not substitute for pooled medians.

Trajectory Validation — LAILA (2010): Held-out seed 0 vs IBTrACS

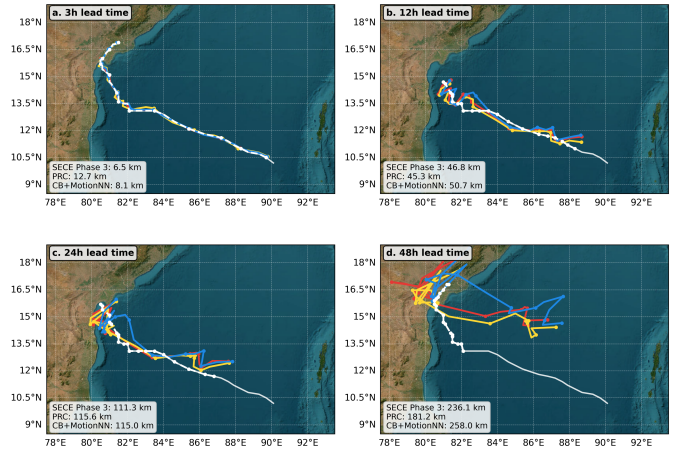


Fig. 4. Cyclone Laila (2010) on held-out seed 0. Each marker is a prediction made from a distinct origin time along the track and offset forward by the panel's lead time, so panels show direct multi-horizon endpoints rather than a recursively rolled-out trajectory. Four panels: 3, 12, 24, and 48 h lead times.

VI. DISCUSSION

The central finding is horizon-dependent. SECE’s routed experts deliver a robust advantage at 3–12 h, where recurvature, coastal proximity, and steering are strongly informative; at 24–48 h, leading tree stacks are statistically indistinguishable [9]–[11], [16]. *Architectural* and *atmospheric* gains peak at different leads: subset fusion helps most at short range, whereas Section V-C shows ERA5 benefit concentrating at 48 h as persistence weakens [7]. The environ expert bridges the two—it repaired 24 h ensemble balance in development (Table I) rather than dominating every per-storm diagnostic.

The leakage correction is part of the claim: test-set-guided phase changes did not replicate on fresh seeds; a frozen design and disjoint held-out tier were required [29]. For the Bay of Bengal rim, at 3–12 h SECE is a defensible default on reanalysis-collocated tree features without a full NWP suite; at 24–48 h, stacking, XGBoost, PRC, and SECE overlap within confidence intervals, so consensus display beats branding one learner dominant. CatBoost-family models lag at short range but add diversity as ensemble members.

Official NIO track verification (IMD/RSMC; 124/202 km mean at 24/48 h in 2009–2013 [14]) uses a different protocol than our IBTrACS hindcast (101/246 km SECE medians); the numbers are not directly comparable, and we rank ML systems under one frozen protocol rather than claiming operational consensus-model skill.

Pre-correction architecture sweeps once appeared to win at every horizon on a single split; that ordering did not replicate on fresh held-out seeds until Phase 3 and the evaluation firewall were frozen [29]. Reporting that failure mode matters for iterative ensemble papers: fusion-path or router choices driven by test-set rank tables can inflate benchmarks even when no learner reads a test label.

For practitioners, the actionable reading is lead-specific. Routed SECE fusion merits priority at 3–12 h, where Bonferroni tests show stable separation from every baseline; at 24–48 h, display consensus among stacking, XGBoost, PRC, and SECE rather than branding one tree learner dominant. Once ERA5 is pre-collocated at track points, the full eight-system confirm remains feasible on modest CPU hardware—relevant where full dynamical suites are capacity-limited along the Bay of Bengal rim [13], [19].

VII. LIMITATIONS

Environmental fields. The locked model uses ERA5 only at the storm centre (14 point variables). We tested $5^\circ/3^\circ$ spatial-patch statistics in development; they did not improve 24–48 h mean error enough to justify adoption (Table I, ~ 0.4 km vs. the environ expert, inside seed noise). Ocean heat content, higher vertical resolution, and forward NWP as distinct from retrospective reanalysis were not incorporated. Gridded or attention-based encoders [7], [20], [28] remain outside this benchmark.

Deterministic outputs. Every system returns a single track point without forecast spread, which matters most at 24–48 h

where Table II shows overlapping leading medians and right-skewed errors; proper probabilistic scores [34] are future work.

Baseline scope. The confirm table is an eight-way tree and hybrid comparison. CNN-GRU and BLSTM were not retrained on the locked ERA5 protocol; Section I summarises why exploratory development under the uniform budget did not justify that cost [11]. Tuned deep encoders on gridded ERA5 remain future work.

Basin and era scope. Training and confirm use North Indian Ocean storms with Bay of Bengal exposure; router menus reflect regional steering (monsoon trough, land proximity). We report no zero-shot or retrained evaluation on the Western North Pacific, Atlantic, or other basins—cross-basin transfer remains future work. Genesis 1940–2024 spans pre-satellite and modern records; although a fixed modern holdout supports the wide window for track error, pre-1979 wind sparsity limits intensity-adjacent interpretation and residual era confounding cannot be fully excluded.

Attribution. Section V-C reports LightGBM *gain* on the environ expert for held-out seed 0 only. Shear and steering components are collinear by construction (shear uses the same wind levels as steering), so rank order within that block is not mechanistic proof; multi-seed SHAP or grouped permutation tests would be needed for stable within-block attribution.

Future work. Three directions could narrow the 24–48 h parity band without relaxing the confirm firewall: physics-constrained displacement models; probabilistic ensemble tracks scored with proper rules [34]; and end-to-end attention over storm history plus gridded ERA5 rather than point collocation [7], [20]. Each should be trained and ranked under the same storm-wise, multi-seed held-out export used here.

VIII. CONCLUSION

We report a leakage-corrected, ERA5-augmented North Indian Ocean benchmark in which SECE Phase 3 wins at 3–12 h with Bonferroni significance, ties leading tree stacks at 24–48 h, and documents rejected spatial patches. Held-out medians are 4.895, 35.538, 101.186, and 246.331 km at 3, 12, 24, and 48 h; SECE is never significantly worse than any evaluated baseline at any horizon, while CatBoost and CB+MotionNN remain the most consistently outperformed comparators at short lead. We document controlled ERA5 ablation, rejected spatial patches, and corrected test-set leakage encountered during our own development so follow-on work can adopt the same storm-wise, multi-seed discipline instead of repeating single-split leaderboard claims. Natural extensions—probabilistic tracks under proper scoring rules, gridded ERA5 encoders, and cross-basin transfer—should be evaluated under an equally explicit confirm firewall [29], [34].

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